Politechnika Warszawska

WYDZIAŁ ELEKTRONIKI
I TECHNIK INFORMACYJNYCH

Wyjaśnialna sztuczna inteligencja

103A-INSZI-MSP-SIWY

DESIGN PROPOSAL

PROJECT'S NO. 3.1.

CAUSAL VALIDATION OF TEXT DETECTOR EXPLANATIONS GENERATED BY LARGE LANGUAGE MODELS

Przyczynowa walidacja wyjaśnień detektorów tekstów generowanych przez duże modele językowe

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Group no. 3

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Number of albums: 347414 | 347990 | 347435

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1. INFORMATION ABOUT PROJECT

We would like to do a research project.

The process and all changes will be documented in our GitHub repository:
https://github.com/KalBia/causal-xai-llm-detectors

2. INTRODUCTION TO THE ISSUE

Detectors of LLM-generated text (systems that classify text as human- or machine-written, typically implemented as a binary classification or a modern ternary classification approach) are commonly explained with **attribution methods** that highlight the tokens driving a decision. But do the highlighted tokens actually influence the detector's verdict? This project validates explanations **causally**: we apply a set of causal interventions to the text and measure whether the features indicated by explanation methods genuinely affect detector decisions.

We also study **explanation washing** — situations where explanations change even though the model's actual decision mechanism has not, which can give a false sense of interpretability.

Research question:

WHICH EXPLANATION METHODS POINT TO FEATURES THAT ARE GENUINELY USED BY LLM-GENERATED-TEXT DETECTORS, AS MEASURED BY CAUSAL INTERVENTIONS ON THE TEXT?

3. PROJECT IMPLEMENTATION DETAILS

3.1. Planned Technology Stack

- ☐ **Language & Environment:** Python ≥ 3.10, dependency management via uv.
- ☐ **Code Quality & Formatting:** ruff.
- ☐ **Version Control:** Git and GitHub (following Conventional Commits standards).
- ☐ **CLI Automation:** Command-line interface scripts built with typer / click, alongside a justfile for build and experiment automation.
- ☐ **Models & Data:** Hugging Face Transformers library for state-of-the-art detectors and curated datasets.
- ☐ **XAI & Causal Interventions:** shap / inseq libraries for feature attribution, paired with a custom module for token ablation and perturbation.
- ☐ **Experiment Tracking:** Weights & Biases (W&B) for logging faithfulness metrics and tracking *explanation washing*.
- ☐ **Configuration Management:** YAML and JSON configuration file

3.2. Planned Software Functionality

- ☐ Development of an evaluation benchmark designed to verify the faithfulness of LLM-generated text detectors.
- ☐ Automated pipeline processing inputs through text perturbation (token ablation or synonym substitution) and evaluating classification argument stability.

3.3. Planned Scope of Experiments

Model Fine-Tuning & Data Split:

- □ Utilizing 3 state-of-the-art detector models (pre-trained + fine-tuned on our dataset divided into train 70%, validation 15%, and test 15%).
- □ Establishing core metrics for training (e.g., F1-score, recall) and explanation methods (SHAP, LIME, Inseq).

Causal Intervention Pipeline (Token Ablation):

- □ On the test set, evaluate text samples (human-written vs. machine-generated) and extract local explanations (e.g., identifying the top-5 most influential words).

- ☐ Automatically execute token ablation by removing or masking the indicated words from the input text.
- ☐ Verify whether the detector's classification label remains unchanged and analyze how its internal reasoning adapts (allowing multiple iterations per sample).

Synonym Perturbation Experiment:

- ☐ Evaluate model robustness and explanation stability by substituting key tokens with synonyms instead of complete removal.

Benchmark & Reporting:

- Generate quantitative tables and visualization charts comparing how explanations shift post-ablation to determine which explanation methods are genuinely faithful and transparent.

4. PROJECT IMPLEMENTATION TIMETABLE

Week	Date Range	Description of Activities	Resources / Time & Compute
1	19 - 25.10.2026	Team formation and topic selection	N/A (conceptual work)
2	26.10 - 01.11.2026	Design proposal: literature review, topic familiarization, task definition, and project scheduling.	N/A (conceptual work)
3	02 - 08.11.2026	Prototype: environment setup, detector models search, acquiring the first detector, dataset sourcing, and metric selection.	Local environment / CPU
4	09 - 15.11.2026	Dataset preparation, detectors fine-tuning, and writing experiment scripts.	GPU (detectors training) Working time: 10-12h
5	16 - 22.11.2026	Detector fine-tuning and writing experiment scripts (cont.).	GPU (detectors training) Training time and experiment prepering: 4-6h
6	23 - 29.11.2026	Implementation of the causal validation mechanism.	Working time: 14h Hardware & Compute: CPU/GPU, unit testing for perturbation scripts
7	30.11 - 06.12.2026	Execution of primary experiments with token ablation and synonymization on the test set.	Working time: 10 h Hardware & Compute: GPU/Multi-core CPU, experiment duration: ~6-8h of continuous computation

8	07-13.12.2026	Documenting experiment workflows, replication, and expanding specific aspects.	Working time: 10h (editorial and analytical work)
9	14-20.12.2026	Analysis of explanation faithfulness plots and detection of *explanation washing*.	Working time: 10h (results analysis)
10	21.12.2026 - 10.01.2027	Results summarization, conclusion drawing, and manuscript formatting.	Conceptual and editorial work
11	10.01 - 14.01.2027	Project submission.	Final verification and submission

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